

SEOUL OBSERVATORY, Korea, Republic of

WMO: 471080

Lat: 37.5714N

Lon: 126.9658E

Elev: 87

StdP: 100.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.5	-23.1	0.5	-8.2	-21.1	0.6	-7.2	7.2	-3.3	6.3	-3.9	2.9	290	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	32.8	24.4	31.4	23.6	30.2	22.7	26.0	30.2	25.2	29.2	24.6	28.4	2.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.1	28.0	24.1	19.2	27.4	23.5	18.4	27.0	81.1	30.3	77.9	29.1	75.3	28.5	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.7	5.1	-14.0	34.7	2.7	1.6	-16.0	35.9	-17.6	36.8	-19.1	37.7	-21.1	38.9		
(5)			-15.3	26.6	2.4	1.1	-17.0	27.4	-18.5	28.1	-19.8	28.7	-21.6	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	-1.9	0.9	6.3	12.9	18.5	23.1	25.7	26.4	21.9	15.5	7.7	0.3	(6)
(7)		DBStd	10.45	4.27	4.15	3.83	3.69	2.77	2.23	2.34	2.53	2.56	3.52	4.48	4.50	(7)
(8)		HDD10.0	1163	369	256	126	13	0	0	0	0	0	5	94	301	(8)
(9)		HDD18.3	2666	627	488	374	167	32	1	0	4	97	318	558		(9)
(10)		CDD10.0	2324	0	1	10	100	264	392	487	510	358	177	26	0	(10)
(11)		CDD18.3	786	0	0	0	3	38	143	229	252	111	11	0	0	(11)
(12)		CDH23.3	6051	0	0	0	22	225	886	1913	2402	576	28	0	0	(12)
(13)		CDH26.7	1777	0	0	0	1	33	219	588	836	99	1	0	0	(13)
(14)	Wind	WSAvg	2.3	2.4	2.5	2.8	2.7	2.5	2.2	2.2	2.1	1.9	2.0	2.2	2.3	(14)
(15)	Precipitation	PrecAvg	1393	19	26	44	83	98	135	396	314	153	51	50	22	(15)
(16)		PrecMax	2356	63	108	124	339	291	497	1131	1238	672	214	165	71	(16)
(17)		PrecMin	761	1	0	3	8	8	16	115	67	5	0	8	3	(17)
(18)		PrecStd	354	16	22	29	65	63	103	208	221	138	41	34	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	13.5	19.0	25.7	29.6	31.8	35.0	35.1	30.4	25.9	19.7	11.7	(19)
(20)			MCWB	5.5	8.1	10.9	15.0	16.8	20.2	25.0	24.7	21.3	17.4	13.6	7.4	(20)
(21)		2%	DB	7.0	10.5	16.2	23.2	27.4	30.3	32.6	33.5	29.0	23.9	17.4	9.8	(21)
(22)			MCWB	3.4	5.4	9.3	14.1	16.6	19.9	24.7	24.6	20.8	16.2	12.3	6.0	(22)
(23)		5%	DB	5.4	8.5	14.1	21.1	25.8	29.1	31.0	32.0	27.8	22.3	15.7	8.2	(23)
(24)			MCWB	2.3	3.9	8.0	12.4	15.9	19.6	24.1	24.3	20.0	15.5	11.3	5.0	(24)
(25)		10%	DB	4.0	6.7	12.2	19.3	24.2	27.8	29.7	30.7	26.6	20.8	14.2	6.6	(25)
(26)			MCWB	1.1	2.8	6.7	11.7	15.3	19.3	23.5	23.8	19.5	14.8	10.1	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	10.0	13.1	17.7	20.2	23.2	27.2	26.9	24.2	19.6	15.5	8.5	(27)
(28)			MCDWB	8.5	11.9	17.3	22.7	24.9	26.5	30.9	30.7	27.7	23.1	18.0	10.6	(28)
(29)		2%	WB	4.2	6.1	10.7	15.6	18.7	22.3	26.1	26.1	23.2	17.9	13.8	6.7	(29)
(30)			MCDWB	6.3	9.5	14.2	20.9	23.3	26.4	30.3	30.2	26.2	21.7	16.4	9.2	(30)
(31)		5%	WB	2.8	4.6	8.8	14.1	17.7	21.5	25.2	25.5	22.1	16.9	12.1	5.2	(31)
(32)			MCDWB	5.0	7.7	12.7	18.8	22.6	26.0	28.9	29.6	25.4	20.5	15.0	7.8	(32)
(33)		10%	WB	1.3	3.3	7.2	12.7	16.8	20.9	24.5	24.9	21.0	15.8	10.6	3.8	(33)
(34)			MCDWB	3.5	6.3	11.3	17.7	21.8	25.3	28.1	29.0	25.0	19.8	13.6	6.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.4	8.1	8.9	9.3	8.1	6.0	6.3	7.7	8.6	7.4	6.8	(35)
(36)			MCDBR	7.5	9.1	10.6	11.6	11.9	10.4	8.0	7.9	9.1	9.8	8.9	7.7	(36)
(37)			MCWBR	5.2	5.5	5.7	5.2	4.5	3.6	2.8	2.7	3.4	4.2	5.4	5.6	(37)
(38)		5% WB	MCDBR	6.6	7.9	9.3	9.4	9.0	7.5	6.2	6.4	6.7	7.8	7.5	7.1	(38)
(39)			MCWBR	5.4	5.8	6.1	5.4	4.8	3.3	2.7	2.7	3.3	4.5	5.9	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.435	0.490	0.573	0.619	0.631	0.702	0.671	0.583	0.510	0.503	0.483	0.433	(40)	
(41)		taud	1.942	1.848	1.710	1.658	1.683	1.606	1.719	1.923	2.045	1.942	1.932	1.971	(41)	
(42)		Ebn at Noon	696	709	691	691	694	645	661	712	742	690	641	664	(42)	
(43)		Edh at Noon	139	174	220	244	242	262	232	185	155	156	141	127	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.50	3.27	4.26	4.97	5.59	5.22	4.07	4.34	4.13	3.58	2.44	2.19	(44)	
(45)		RadStd	0.16	0.26	0.42	0.35	0.52	0.54	0.62	0.61	0.45	0.25	0.24	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.17	N/A	N/A	N/A	N/A	N/A	+95	+63
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+50

Nomenclature: See separate page